

Unit 1 Review Packet

Foundations of Algebra

Slope • Linear Equations & Forms • Graphing • Arithmetic Sequences • Functions, Domain & Range • Key Features

Key Ideas

Slope: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$

Slope-intercept: $y = mx + b$

Point-slope: $y - y_1 = m(x - x_1)$

Standard form: $Ax + By = C$

x-intercept: let $y = 0$, solve for x

y-intercept: let $x = 0$, solve for y

Explicit rule: $a_n = a_1 + d(n - 1)$

Recursive rule: $a_1 = \text{first term}, a_n = a_{n-1} + d$

Domain = inputs (x -values)

Range = outputs (y -values)

$f(3)$ means "substitute $x = 3$ "

$[]$ includes ($\leq, \geq$); $()$ excludes ($<, >$)

∞ and $-\infty$ always get $()$

Worked Example

A line passes through $(-2, 3)$ and $(4, 0)$. Write its equation in slope-intercept form, find both intercepts, write it in standard form, and graph it.

Step 1 – Find the slope.

$$m = \frac{0 - 3}{4 - (-2)} = \frac{-3}{6} = -\frac{1}{2}$$

Step 2 – Point-slope, then solve for y . Use $(-2, 3)$:

$$y - 3 = -\frac{1}{2}(x - (-2)) \Rightarrow y - 3 = -\frac{1}{2}x - 1 \Rightarrow y = -\frac{1}{2}x + 2$$

Step 3 – Intercepts.

y -int: let $x = 0 \Rightarrow y = 2$, so $(0, 2)$.

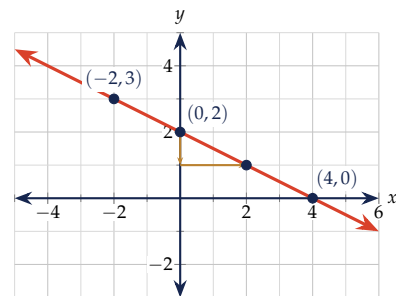
x -int: let $y = 0 \Rightarrow 0 = -\frac{1}{2}x + 2 \Rightarrow x = 4$, so $(4, 0)$.

Step 4 – Standard form. Multiply every term by 2:

$$2y = -x + 4 \Rightarrow x + 2y = 4$$

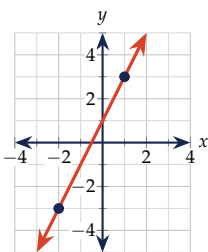
Step 5 – Graph. Plot $(0, 2)$, then use the slope $-\frac{1}{2}$: down 1, right 2. Check with the table.

x	-2	0	2	4
y	3	2	1	0



Part A – Slope & Linear Equations

1. Find the slope of the line.



$m =$ _____

2. Find the slope of the line that passes through $(3, -4)$ and $(7, 8)$.

$m =$ _____

3. Find the slope of the line represented by the table.

x	2	4	6	8
y	9	5	1	-3

$m =$ _____

4. Identify the slope and y -intercept of

$$y = -\frac{3}{4}x + 6$$

$m =$ _____ $b =$ _____

5. Find the x - and y -intercepts of

$$3x + 4y = 24$$

x -int: _____ y -int: _____

6. Find the y -intercept of

$$4x - 5y = 20$$

Answer: _____

7. Rewrite in slope-intercept form:

$$6x + 3y = 15$$

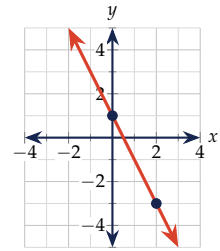
Answer: _____

8. Rewrite in slope-intercept form:

$$-8x - 2y = 10$$

Answer: _____

9. Write the equation of the line shown.



Answer: _____

10. Write the equation of the line through $(-3, 5)$ with slope 2. Give your answer in slope-intercept form.

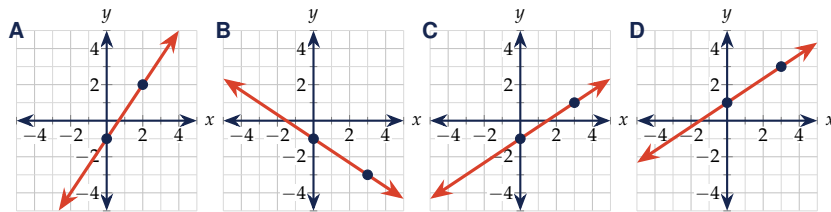
Answer: _____

11. Write the equation, in slope-intercept form, of the line through $(2, 7)$ and $(-4, -2)$.

Answer: _____

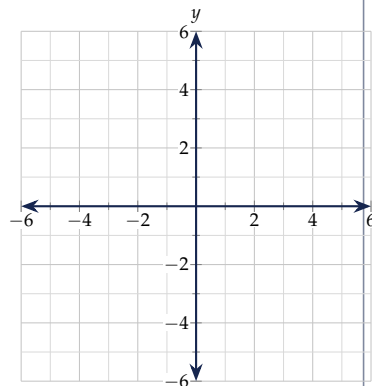
12. Which graph represents $y = \frac{2}{3}x - 1$?

Answer: _____



13. Graph the line $3x - 2y = 6$. Find the intercepts *or* convert to slope-intercept form first — show your method.

Work:



14. A phone plan charges a \$20 activation fee plus \$35 per month, modeled by $y = 35x + 20$. After how many months will the total cost reach \$300?

Answer: _____

15. A plumber charges \$55 per hour plus a flat fee. A 4-hour job costs \$260.

(a) Write an equation for the total cost y of an x -hour job.

Answer: _____

(b) What is the cost of a 6-hour job?

Answer: _____

Part B – Arithmetic Sequences

16. The 5th term of an arithmetic sequence is 11. The common difference is 4. What is the 12th term?

Answer: _____

17. Find the 50th term of the sequence

8, 14, 20, 26, ...

Answer: _____

18. Write a *recursive* rule for the sequence

5, 13, 21, 29, ...

$a_1 =$ _____, $a_n =$ _____

19. Given the sequence 14, 10, 6, 2, -2, ...

(a) Write an explicit (closed) rule for the sequence.

Answer: _____

(b) Find the 100th term.

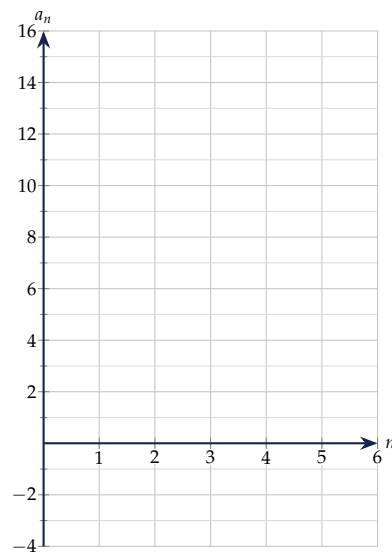
Answer: _____

(c) Complete the table and graph the first 5 terms. Should the points be connected (continuous) or not (discrete)? _____

n	1	2	3	4	5
a_n					

(d) Domain of the first 5 terms: _____

Range of the first 5 terms: _____



20. Given the recursive rule $a_1 = -3$, $a_n = a_{n-1} + 5$, find the first 5 terms of the sequence.

Answer: _____

21. At the skating rink, Maya pays \$8 to rent skates and then \$4 per hour. How many hours can she skate if she wants to spend exactly \$32?

Answer: _____

Part C – Functions, Domain & Range, Key Features

22. If $f(x) = 2x + 7$ and the domain is $\{-3, 0, 5\}$, what is the range?

Answer: _____

23. Given $f(x) = x^2 - 3x + 4$, find $f(-2)$.

Answer: _____

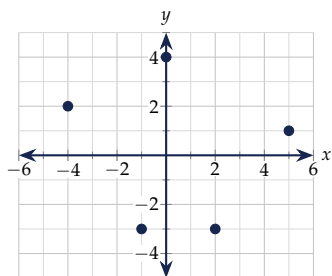
24. Given $f(x) = 4x - 9$, find the value of x when $f(x) = 15$.

Answer: _____

25. Find the domain and range of the discrete points.

Domain:

Range:

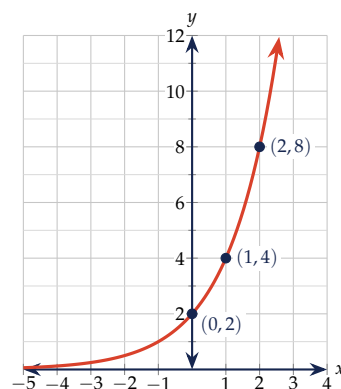


26. Use the graph of f to evaluate.

(a) $f(1) =$ _____

(b) $f(0) =$ _____

(c) If $f(x) = 8$,
 $x =$ _____



27. Convert to interval notation.

(a) $x > -3$ _____ (b) $x \leq 7$ _____

28. Convert to inequality notation.

(a) $[-2, 9)$ _____ (b) $(-4, 6]$ _____

29. Use the graph of f to answer each question. (The arrow means the graph keeps going.)

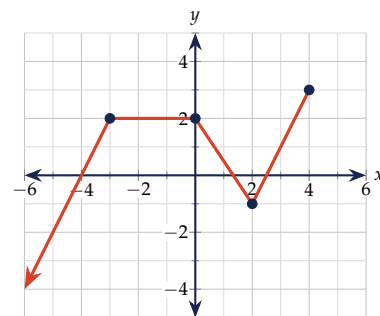
(a) $f(-3) =$ _____ (b) $f(2) =$ _____

(c) Domain: _____ (d) Range: _____

(e) Interval(s) of increase: _____

(f) Interval(s) of decrease: _____

(g) End behavior: as $x \rightarrow -\infty$, $y \rightarrow$ _____



30. Find the key features of the graph. (Closed dot = endpoint; arrow = keeps going.)

Domain:	Range:
x-intercept:	y-intercept:
Interval of increase:	Interval of decrease:
End behavior: as $x \rightarrow \infty$, $y \rightarrow$ _____	

